

2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter using MATLAB

```
clc;

clear;

close all;

% Specifications

Rp = 1; % Passband Ripple (dB)

Rs = 40; % Stopband Attenuation (dB)

Wp = 0.5; % Passband Edge

Ws = 0.35; % Stopband Edge

% Find Filter Order

[n,Wn] = cheblord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter

[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response

figure;

freqz(b,a);

title('Frequency Response');

% Impulse Response

figure;

impz(b,a);

title('Impulse Response');

% Group Delay

figure;

grpdelay(b,a);

title('Group Delay');

% Pole-Zero Plot

figure;

zplane(b,a);
```

```

title('Pole-Zero Plot');

fprintf('Filter Order = %d\n',n);

fprintf('Cutoff Frequency = %.4f\n',Wn);

```



